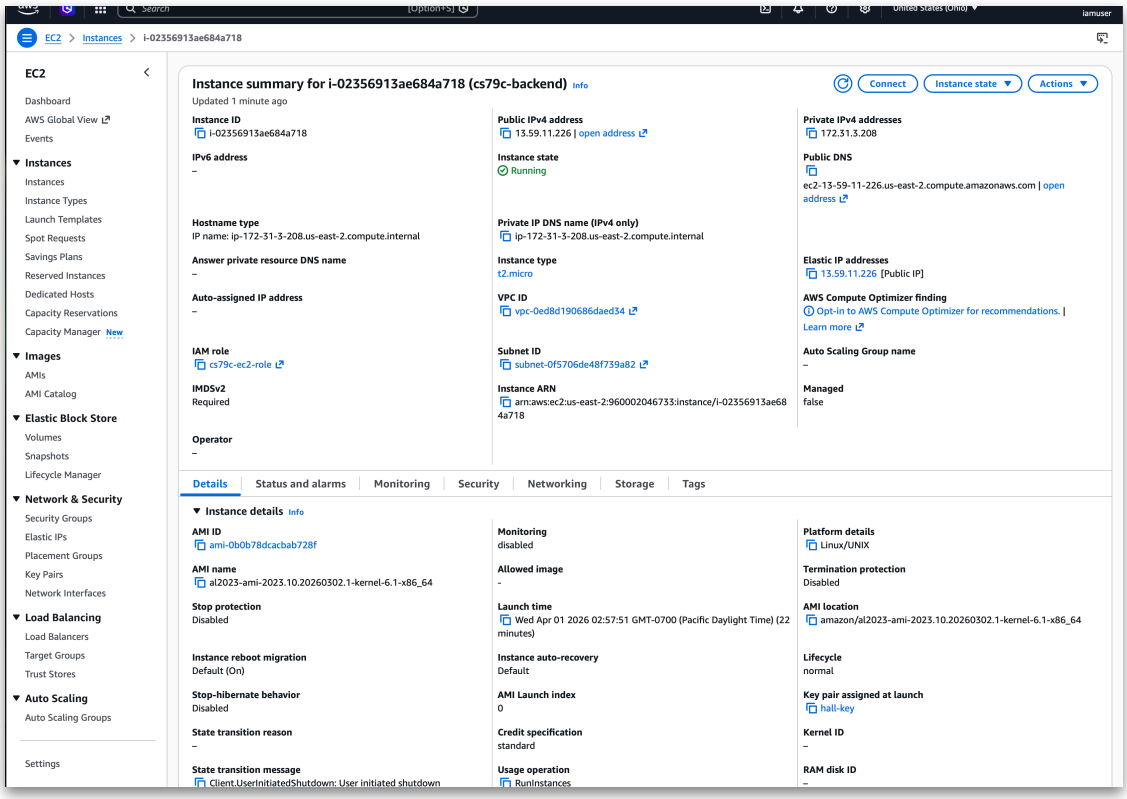
AWS Cloud Stack for the AWS Learning Portal

Project: AWS Learning Portal
Course: CS 79C Compute Engines in Amazon Web Services
Instructor: Vicky Tanya Seno
Student: Siarhei Hancharou
Student ID: 1712666

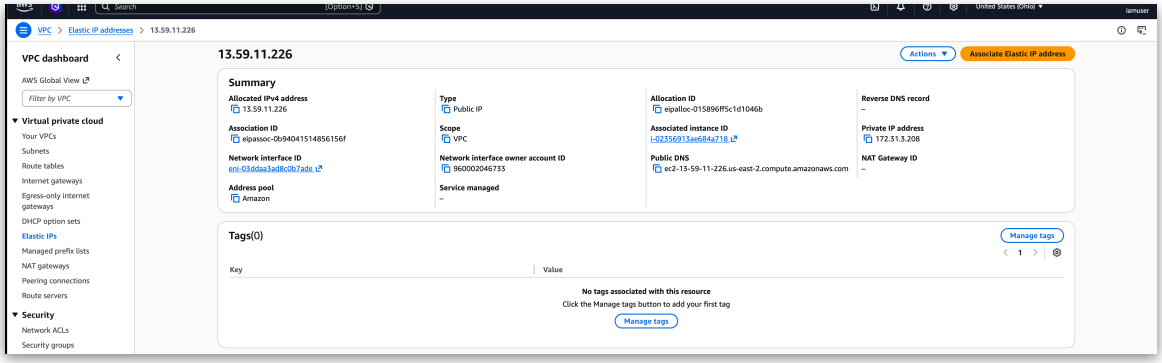
1. Core AWS Services

Compute & Application Layer:
- Amazon EC2 - hosting frontend and backend

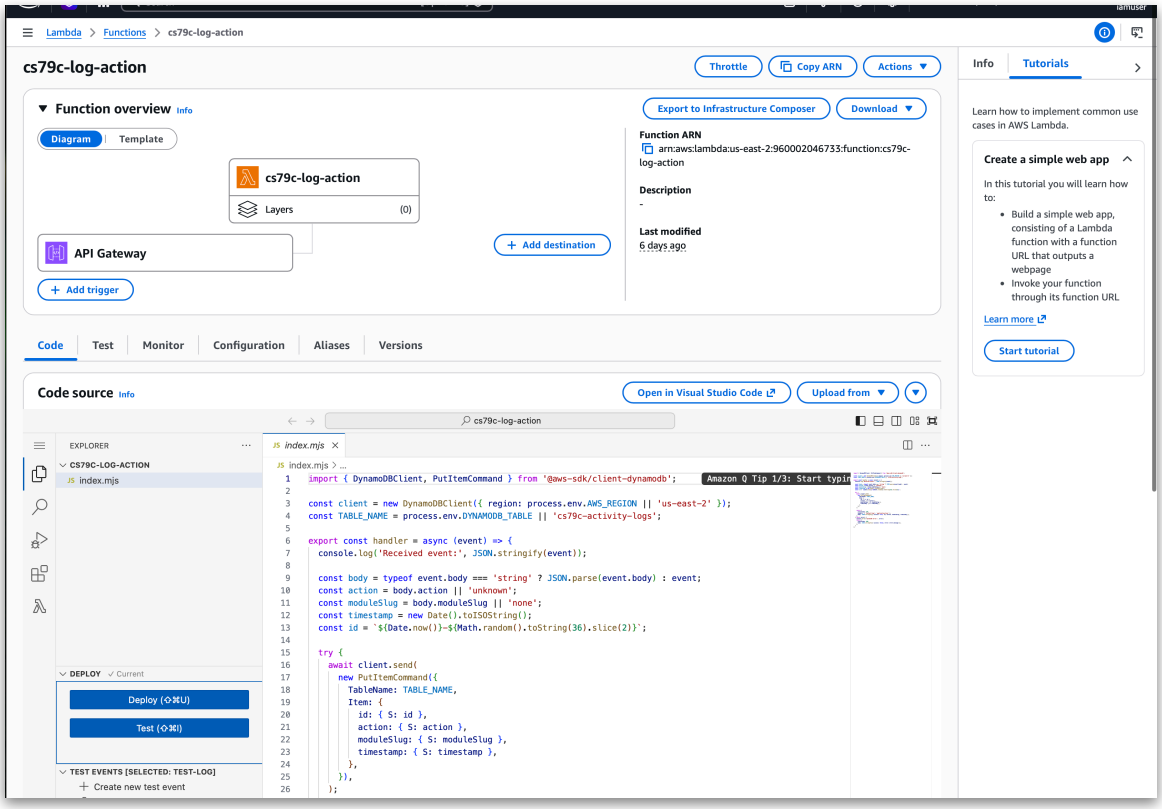




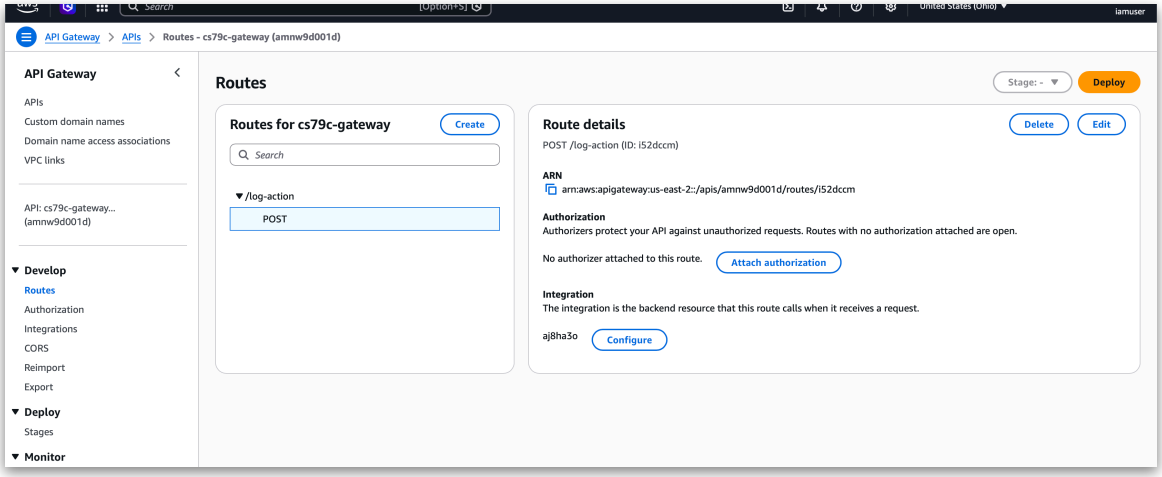
- Configured a static Public IPv4 13.59.11.226 using an Elastic IP for stable public access



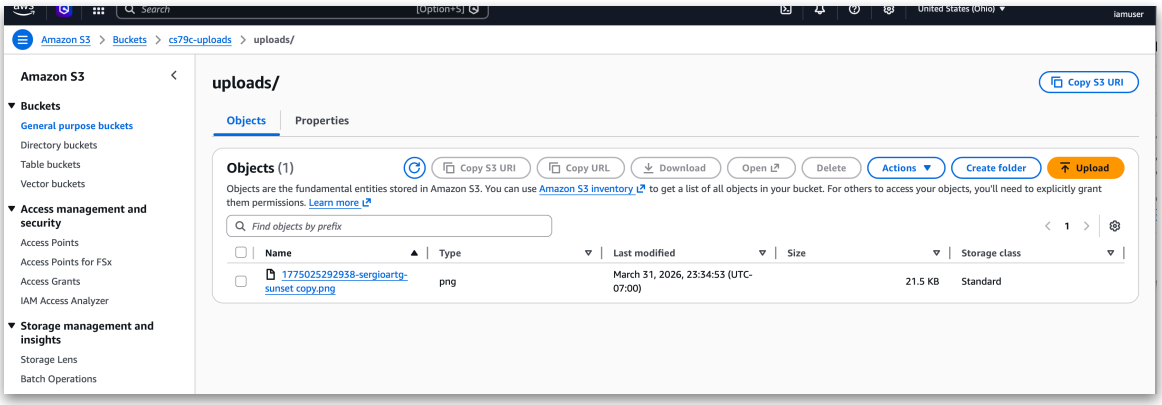
- AWS Lambda - serverless processing



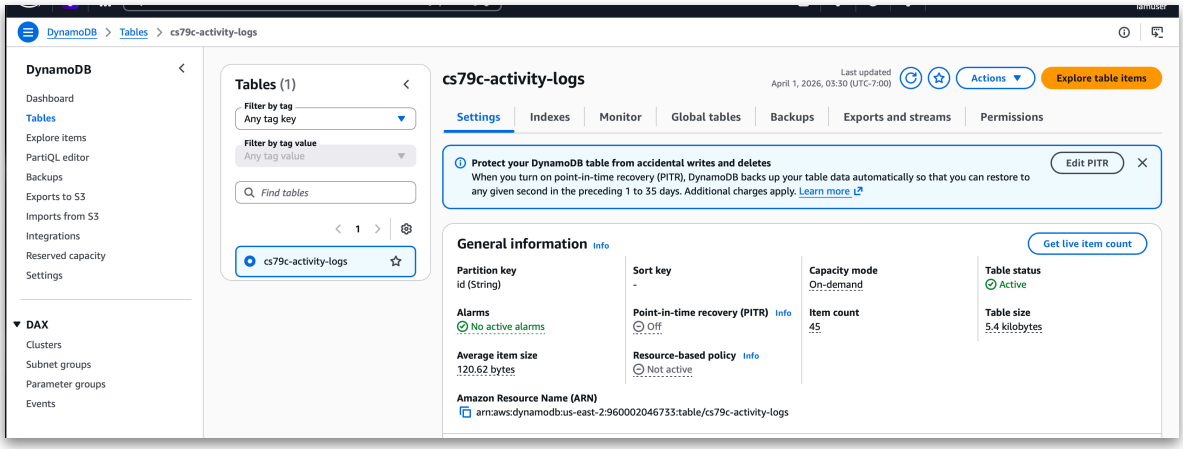
- Amazon API Gateway - API routing



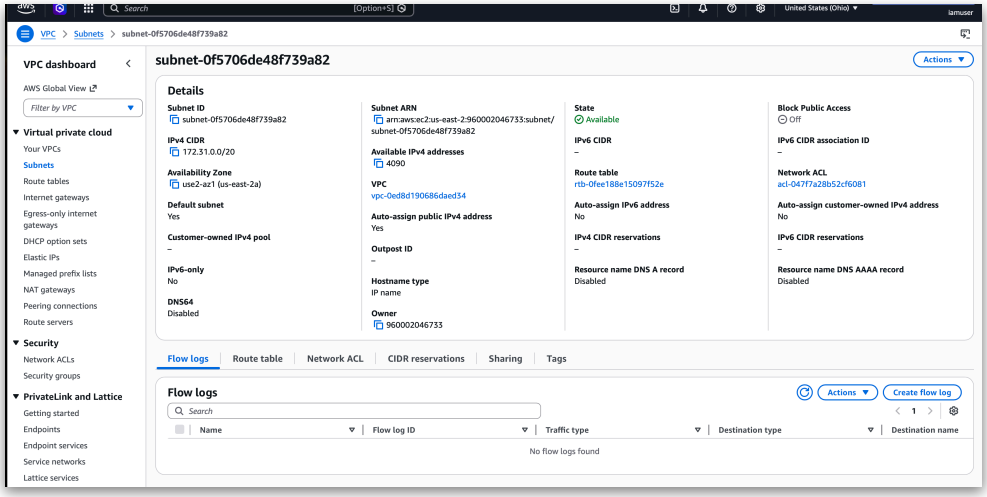
Storage & Database:
- Amazon S3 - file storage



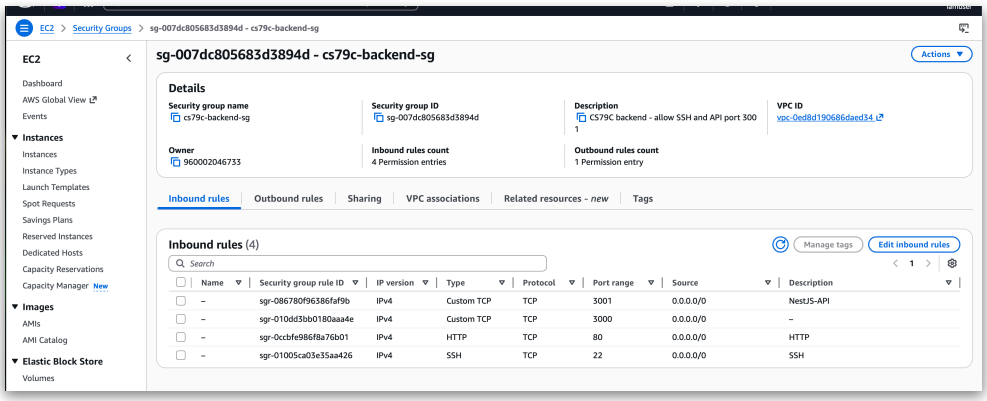
- Amazon DynamoDB - NoSQL database for logs



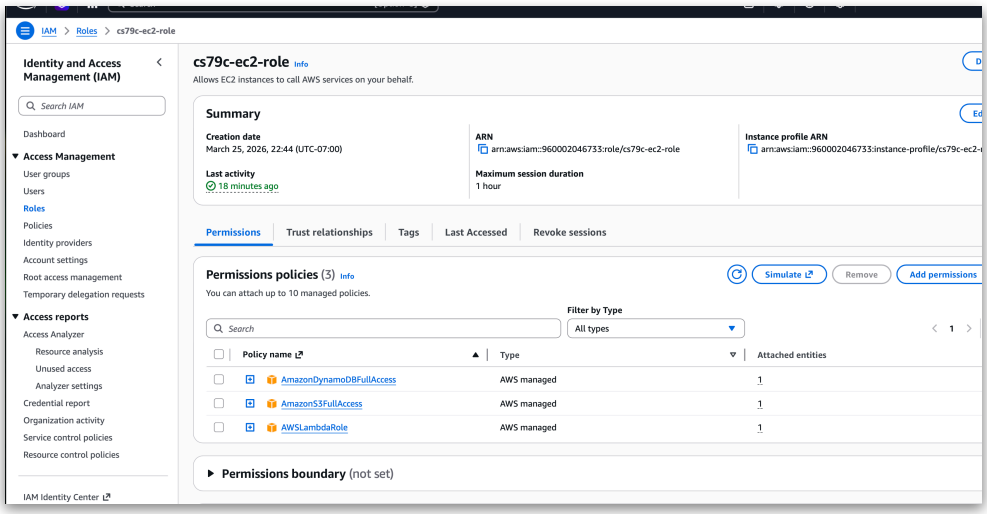
Networking & Security:
- Amazon VPC (default VPC networking for the current EC2 deployment)
- Public Subnets (Multi-AZ)



- Security Groups

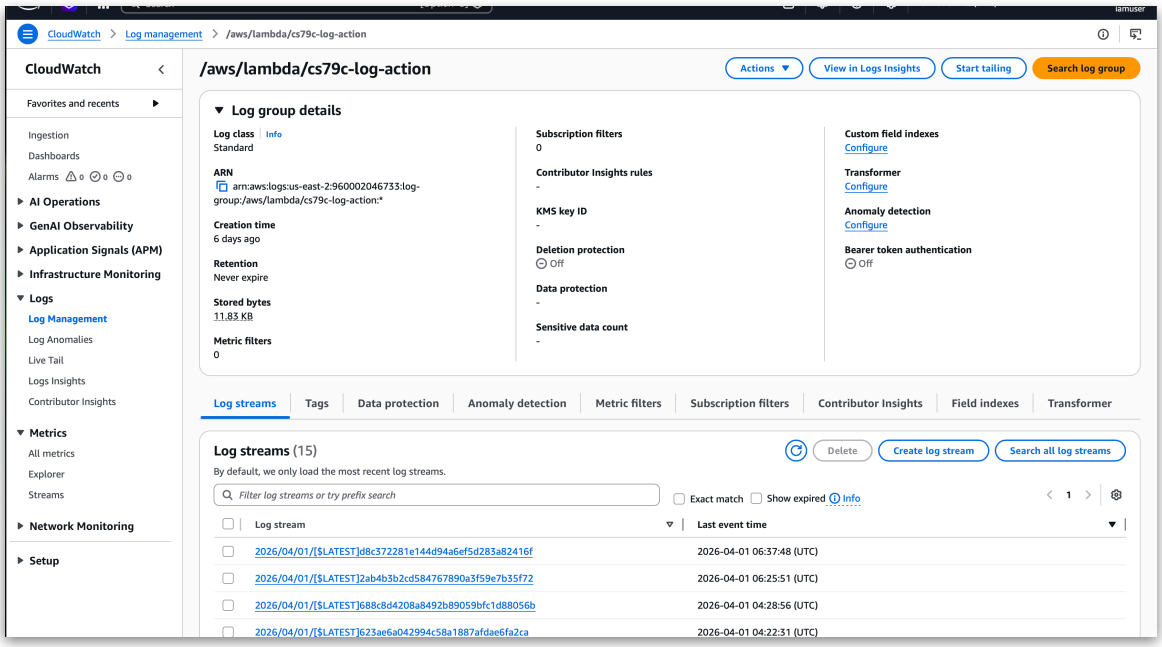


- IAM Roles / Instance Profiles

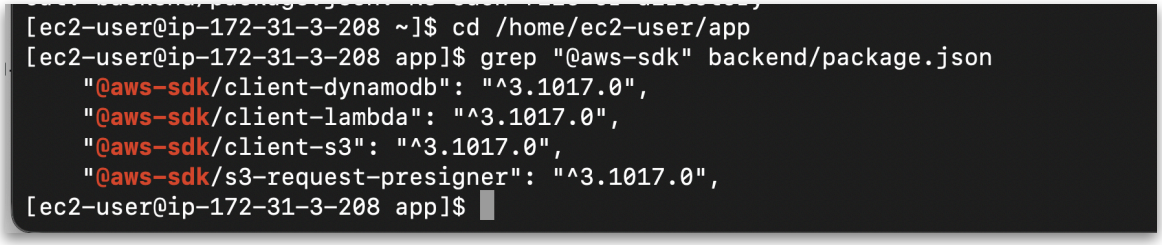


2. Monitoring & DevOps

- Amazon CloudWatch - monitoring and logging



- AWS SDK for JavaScript v3 - integration layer



3. Operating System

- Amazon Linux 2023

4. Architecture Summary

Client → EC2 (Frontend)
Frontend → Backend (EC2)
Backend → S3
Backend → API Gateway → Lambda
Lambda → DynamoDB
CloudWatch → Monitoring